OMEGA

OMEGA operates in the electronic devices manufacturing industry. Its flagship product is the ALFA device, which is sold at a price of €900 per unit, with avoidable (variable) production costs equal to €450 per unit.

In 2024, the management evaluates a new investment opportunity. OMEGA could purchase a new machine to produce a next-generation device called BETA, which would be sold at €1,200 per unit, with avoidable production costs equal to €550 per unit.

The expected sales forecasts for the four-year period 2025–2028 are reported below. It should be noted that the introduction of BETA reduces the demand (and therefore the production) of ALFA.

Year	2025	2026	2027	2028
Demand BETA	1,200	1,300	1,400	1,500
Reduction Demand ALFA	1,300	1,400	1,500	1,600

To produce BETA, OMEGA must purchase a new machine at a cost of €400,000.
The machine has a depreciation life of 5 years, with straight-line depreciation. The acquisition date was at December 31st, 2024, and in January 1, 2025 the machine becomes immediately operational. At the end of 2028 the machine is sold for €40,000.

When evaluating the investment, the following additional effects must also be considered:

1. In addition to the current workforce, one additional worker must be hired, with an annual salary of €30,000.
2. Training costs of €15,000 per year will be incurred in 2025 and 2026.
3. Additional administrative costs will be incurred as follows:
    - €8,000 in 2025
    - €10,000 in 2026
    - €12,000 in 2027 and 2028
4. The company obtains a subsidized loan from the Ministry of Industry for €100,000, at an interest rate of 5%, starting January 1, 2025. The loan is repaid entirely at the end of 2028.

Assume the following:

- Corporate tax rate: 30%
- Cost of equity: 25%
- There are no changes in working capital.

Questions

1. Compute the Net Present Value (NPV) of the project using the FCFF approach.
2. Compute the Net Present Value (NPV) using the FCFE approach.